SDEV 2401

Rapid Backend App Development

Management Commands - 1

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Expectations - What I expect from you

- No Late Assignments

- No Cheating

- Be a good classmate

- Don’t waste your time

- Show up to class

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Agenda

On the right is what we will cover today.

- What Are Django Management Commands?
- Setting Up a Custom Command
- Example – Archiving Old Blog Posts
- Adding Arguments to Commands
- Sending Notifications with a Command
- Summary, Tips and Best Practices

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What Are Django Management Commands?

- Django provides built-in commands like `runserver`, `migrate`, and `createsuperuser`.
- You can create your own commands to automate project-specific tasks.
- Custom commands are run with `python manage.py <command_name>`.

Example Use Cases:

- Bulk updating user statuses.
- Generating reports.
- Cleaning up old data.
- Sending periodic emails.
- Data migrations or transformations.

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Setting Up a Custom Command

Here's how to create a custom management command in Django in an app called `myapp`.

Directory Structure:

```
myapp/
  management/
    __init__.py
    commands/
      __init__.py
      archive_old_posts.py
```

Key Points:

- Place your command in `management/commands/` inside your app.
- The filename becomes the command name (`archive_old_posts`).
- Each command is a Python module with a `Command` class.

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Example – Archiving Old Blog Posts

```python
from django.core.management.base import BaseCommand
from blog.models import Post
from django.utils import timezone
from datetime import timedelta

class Command(BaseCommand):
    help = 'Archive blog posts older than 1 year'

    def handle(self, *args, **kwargs):
        cutoff = timezone.now() - timedelta(days=365)
        count = Post.objects.filter(published_date__lt=cutoff, archived=False).update(archived=True)
        self.stdout.write(self.style.SUCCESS(f'Archived {count} posts.'))
```

- This command marks posts older than a year as archived.
- Uses Django ORM to filter and update records.
- Prints a success message with the number of posts archived.

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Adding Arguments to Commands

Arguments are useful for making commands more flexible.

```
def add_arguments(self, parser):
    parser.add_argument('--days', type=int, default=30, help='Archive posts older than N days')
```

Usage:

```
python manage.py archive_old_posts --days 90
```

- `add_arguments` lets you customize command-line options.
- Here, `--days` lets users specify the age threshold for archiving.
- Default values and help text improve usability.

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Sending Notifications with a Command

- This command checks for unreviewed comments and emails the admin if any exist.
- Demonstrates integrating Django's email system in a management command.
- Useful for periodic checks via cron or task schedulers.

```
from django.core.mail import send_mail

class Command(BaseCommand):
    help = 'Notify admins of unreviewed comments'

    def handle(self, *args, **kwargs):
        from blog.models import Comment
        unreviewed = Comment.objects.filter(reviewed=False)
        if unreviewed.exists():
            send_mail(
                'Unreviewed Comments',
                f'There are {unreviewed.count()} unreviewed
                'admin@mysite.com',
                ['admin@mysite.com']
            )
            self.stdout.write(self.style.SUCCESS('Notificat
        else:
            self.stdout.write('No unreviewed comments.')
```

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Summary, Tips and Best Practices

- Use management commands for tasks that need project context (models, settings).
- Add helpful messages and error handling for better UX.
- Test commands in a development environment before running in production.
- Schedule commands with cron or Celery Beat for automation.

Summary: Custom management commands are powerful tools for automating, maintaining, and extending your Django projects beyond the web interface.

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Example

Let’s go do an example together